

## 3.3 Linear transformations and marginals

### 3.3.1 Affine maps

#### Theorem 3.3.1 — Affine transformations

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , let  $\mathbf{A}$  be any  $m \times n$  matrix and  $\mathbf{b} \in \mathbb{R}^m$ . Then

$$\mathbf{AY} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

*Proof.* For  $\mathbf{c} \in \mathbb{R}^m$ ,  $\mathbf{c}^\top(\mathbf{AY} + \mathbf{b}) = (\mathbf{A}^\top \mathbf{c})^\top \mathbf{Y} + \mathbf{c}^\top \mathbf{b}$  is a linear combination of  $\mathbf{Y}$  plus a constant, which is normal by Definition 3.1.1 and Lemma 3.1.1(c). So  $\mathbf{AY} + \mathbf{b}$  is normal, and its mean and covariance are given by Theorem 2.2.1. ■

The proof is one line because the definition was chosen to make it one line. There is no rank condition on  $\mathbf{A}$ . If  $\mathbf{A}$  has more rows than  $\text{rank}(\boldsymbol{\Sigma})$ , the result is a singular normal vector, and that is allowed. With a density-based definition, the same theorem needs  $\mathbf{A}$  to have full row rank and  $\boldsymbol{\Sigma}$  to be positive definite, and the most useful cases fall outside it. The moment generating function gives a second proof:  $E e^{\mathbf{t}^\top(\mathbf{AY} + \mathbf{b})} = e^{\mathbf{t}^\top \mathbf{b}} M_{\mathbf{Y}}(\mathbf{A}^\top \mathbf{t}) = \exp\{\mathbf{t}^\top(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{t}^\top \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top \mathbf{t}\}$ .

#### Corollary 3.3.2 — Marginal distributions

Every subvector of a normal vector is normal. If  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and  $\mathbf{Y}_1$  consists of the entries with indices in a set  $I$ , then  $\mathbf{Y}_1 \sim N(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_{II})$ , where  $\boldsymbol{\mu}_I$  and  $\boldsymbol{\Sigma}_{II}$  keep the entries, rows and columns indexed by  $I$ . In particular  $Y_i \sim N(\mu_i, \sigma_{ii})$ .

*Proof.*  $\mathbf{Y}_1 = \mathbf{AY}$ , where the rows of  $\mathbf{A}$  are the coordinate vectors  $\mathbf{e}_i^\top$ ,  $i \in I$ . Then  $\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}_I$  and  $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \boldsymbol{\Sigma}_{II}$ . ■

#### Corollary 3.3.3 — Standardization and rotation

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ .

- (a) If  $\boldsymbol{\Sigma}$  is positive definite and  $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$  with  $\mathbf{A}$  square, then  $\mathbf{A}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim N_n(\mathbf{0}, \mathbf{I})$ . Its entries are independent standard normal variables.
- (b) If  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  and  $\mathbf{Q}$  is an  $n \times n$  orthogonal matrix, then  $\mathbf{Q}^\top \mathbf{Y} \sim N_n(\mathbf{Q}^\top \boldsymbol{\mu}, \sigma^2 \mathbf{I})$ .

*Proof.* Both follow from Theorem 3.3.1:  $\mathbf{A}^{-1}\mathbf{A}\mathbf{A}^\top(\mathbf{A}^{-1})^\top = \mathbf{I}$  and  $\mathbf{Q}^\top(\sigma^2 \mathbf{I})\mathbf{Q} = \sigma^2 \mathbf{I}$ . Independence in (a) was shown in the proof of Proposition 3.2.3. ■

Part (b) is the probabilistic basis of the canonical form of the linear model. Spherical normal errors look the same in every orthonormal coordinate system, so one may rotate to coordinates adapted to  $\mathcal{C}(\mathbf{X})$ , in which fitted values and residuals occupy separate independent coordinates. [Chapter 4](#) uses this rotation to find the distributions of sums of squares.

### 3.3.2 The rank of a normal vector

Theorem 3.1.2 writes a normal vector as an affine function of independent standard normals *in distribution*. When  $\Sigma$  is singular the representation can be made exact, with as few standard normals as the rank.

#### Proposition 3.3.4

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  with  $\text{rank}(\Sigma) = r \geq 1$ , and let  $\Sigma = \mathbf{Q}\Lambda\mathbf{Q}^\top$ , where  $\mathbf{Q}$  is  $n \times r$  with orthonormal columns and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$  holds the positive eigenvalues. Put  $\mathbf{W} = \Lambda^{-1/2}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$ . Then  $\mathbf{W} \sim N_r(\mathbf{0}, \mathbf{I}_r)$ , and

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{Q}\Lambda^{1/2}\mathbf{W} \quad \text{with probability one.}$$

*Proof.* By [Theorem 3.3.1](#),  $\mathbf{W}$  is normal with covariance  $\Lambda^{-1/2}\mathbf{Q}^\top\mathbf{Q}\Lambda\mathbf{Q}^\top\mathbf{Q}\Lambda^{-1/2} = \mathbf{I}_r$ . Next,  $\mathbf{Q}\Lambda^{1/2}\mathbf{W} = \mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$ , and  $\mathbf{Q}\mathbf{Q}^\top$  is the symmetric idempotent matrix with range  $\mathcal{C}(\Sigma)$  (the orthogonal projection onto it, in the language of [Chapter 6](#)). Since  $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$  with probability one (Theorem 2.2.3),  $\mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{Y} - \boldsymbol{\mu}$  with probability one. ■

So a normal vector of rank  $r$  is exactly  $r$  independent standard normals placed in  $\mathbb{R}^n$  by an injective affine map. Exercise (B4) gives the same construction for any random vector, with uncorrelated entries in place of independent normal ones. Singular normal vectors are lower-dimensional normal vectors in disguise.

#### Example. (Least squares under normal errors)

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\mathbf{X}$  of full column rank  $p$ , and let  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ , the symmetric idempotent matrix of rank  $p$  with  $\mathbf{M}\mathbf{X} = \mathbf{X}$  ([Chapter 5](#) and [Chapter 6](#)). By [Theorem 3.3.1](#):

- (a)  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$ , a nonsingular normal vector;
- (b)  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\lambda}^\top\boldsymbol{\beta}, \sigma^2\boldsymbol{\lambda}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\boldsymbol{\lambda})$  for every  $\boldsymbol{\lambda} \in \mathbb{R}^p$ ;
- (c) the fitted values  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{M})$ , singular of rank  $p$ ;
- (d) the residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ , singular of rank  $n - p$ .

No density for  $\hat{\mathbf{Y}}$  or  $\hat{\boldsymbol{\varepsilon}}$  was needed, and none exists.

### 3.3.3 Independent normal vectors

**Lemma 3.3.5**

If  $\mathbf{Y}_1 \sim N_{n_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$  and  $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$  are independent, then  $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$  is jointly normal with mean  $(\boldsymbol{\mu}_1^\top, \boldsymbol{\mu}_2^\top)^\top$  and block-diagonal covariance  $\text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$ . Consequently, if also  $n_1 = n_2$ , then  $\mathbf{Y}_1 + \mathbf{Y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$ .

*Proof.*  $\mathbf{a}_1^\top \mathbf{Y}_1 + \mathbf{a}_2^\top \mathbf{Y}_2$  is a sum of independent normal variables, which is normal by Lemma 3.1.1(c). Independence makes the cross-covariance zero. The sum is  $[\mathbf{I}, \mathbf{I}]$  applied to the stacked vector. ■

Independence is essential in this lemma. Two normal vectors defined on the same probability space need not be jointly normal, as we now see.

**3.3.4 Normal marginals are not enough**

Corollary 3.3.2 cannot be reversed. A vector whose entries are each normal need not be jointly normal. Two examples make the point, one extreme and one realistic.

**Example. (A random sign)**

Let  $Z \sim N(0, 1)$  and let  $S$  be independent of  $Z$  with  $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$ . Put  $W = SZ$ . By symmetry of  $Z$ ,  $W \sim N(0, 1)$ . Also  $\text{Cov}(Z, W) = E(S)E(Z^2) = 0$ . But  $Z + W = (1 + S)Z$  equals zero with probability  $\frac{1}{2}$  and is otherwise  $N(0, 4)$ . A variable with an atom at zero and a continuous part is not normal, so  $(Z, W)$  is not jointly normal. And  $Z$  and  $W$  are far from independent:  $|W| = |Z|$  always.

**Example. (A mixture of two correlations)**

Fix  $0 < \rho < 1$ . With probability  $\frac{1}{2}$  draw  $(Y_1, Y_2)$  from the bivariate normal with unit variances and correlation  $\rho$ , and otherwise from the one with correlation  $-\rho$ . Both components have  $N(0, 1)$  margins, so the mixture does too, exactly. The covariance is  $\frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0$ . The pair is nevertheless dependent: since  $E(Y_1^2 Y_2^2) = 1 + 2\rho^2$  for a standard bivariate normal with correlation  $r$ , the squares have correlation  $\rho^2$ , not zero. The direction  $\mathbf{a} = (1, 1)^\top / \sqrt{2}$  exposes the failure of joint normality.  $S = (Y_1 + Y_2) / \sqrt{2}$  is an equal mixture of  $N(0, 1 + \rho)$  and  $N(0, 1 - \rho)$ . It has variance 1 but kurtosis  $E S^4 = \frac{3}{2}\{(1 + \rho)^2 + (1 - \rho)^2\} = 3(1 + \rho^2)$ , where a normal variable would have 3.

Figure 3.3.1 shows a simulation with  $\rho = 0.8$  and 200,000 draws. Kolmogorov–Smirnov tests of normality for the two margins give  $p$ -values 0.84 and 0.29. The sample correlation is  $-0.0033$ , the correlation of the squares is 0.640 (theory 0.64), and the kurtosis of  $S$  is 4.88 (theory 4.92). The Kolmogorov–Smirnov distance of  $S$  from  $N(0, 1)$  is 0.067, which with this many draws rejects normality at any conventional level.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(302)
```

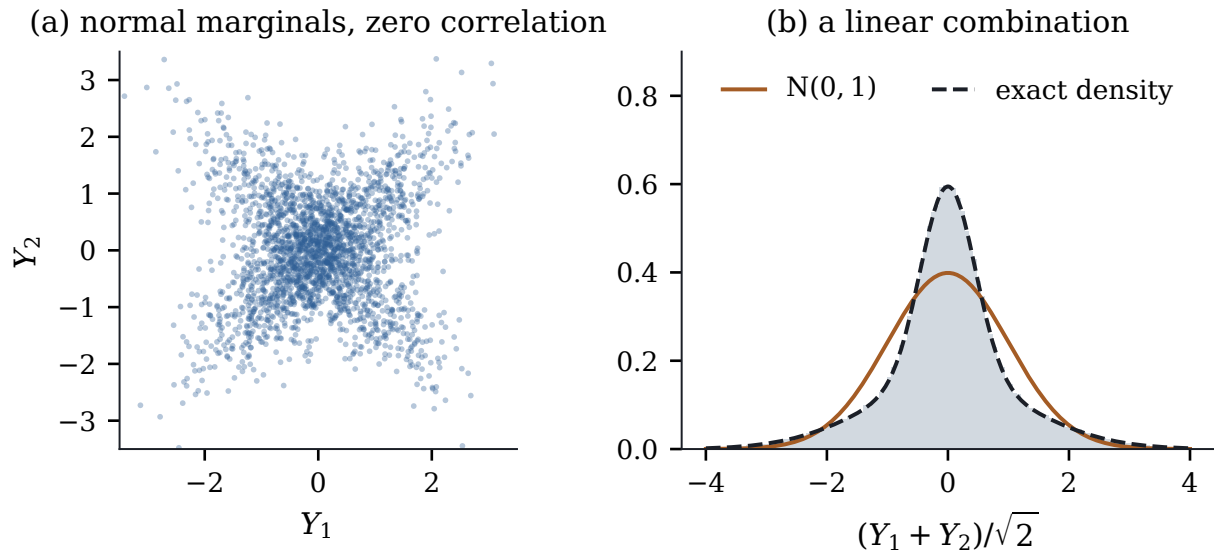


Figure 3.3.1: An equal mixture of bivariate normals with correlations  $\pm 0.8$ . (a) Both margins are exactly  $N(0, 1)$  and the correlation is zero, but the scatter is an X, not an ellipse. (b) The linear combination  $(Y_1 + Y_2)/\sqrt{2}$  has variance one but is sharply peaked and heavy-tailed compared with  $N(0, 1)$ . By Definition 3.1.1, one non-normal direction is enough to rule out joint normality.

```
rho, N = 0.8, 200_000

sign = rng.choice([-1.0, 1.0], size=N) # which component each draw uses
Z1, Z2 = rng.standard_normal((2, N))
Y1 = Z1
Y2 = sign * rho * Z1 + np.sqrt(1 - rho**2) * Z2 # corr(Y1, Y2 | sign) = sign * rho

S = (Y1 + Y2) / np.sqrt(2) # a direction that is not normal
print("marginal KS p-values:", stats.kstest(Y1, "norm").pvalue, stats.kstest(Y2, "norm").pvalue)
print("correlation:", np.corrcoef(Y1, Y2)[0, 1])
print("corr of squares:", np.corrcoef(Y1**2, Y2**2)[0, 1])
print("kurtosis of S:", stats.kurtosis(S, fisher=False), " theory:", 3 * (1 + rho**2))
print("KS p-value of S:", stats.kstest(S, "norm").pvalue)
```

[Run this code online](#)

*Idea.* Normality of a vector is a property of *every direction*, not of the coordinate axes. Normal margins with zero correlation can hide strong dependence. The theorems of the next two sections, which turn zero covariance into independence and give linear conditional means, need *joint* normality. They are not about variables that merely happen to be normal one at a time.

## 3.3.5 Exercises

### 3.3.5.1 A. Check your understanding

**Exercise A1.**

Let  $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\mu} = (0, 1, 2)^\top$  and  $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ . Find the joint distribution of  $U = Y_1 + Y_2 + Y_3$ ,  $V = Y_1 - Y_3$  and  $W = Y_1 - 2Y_2 + Y_3$ . Which of the pairs are independent? Is  $V$  independent of  $(U, W)$ ?

**Exercise A2.**

In [Example \(Least squares under normal errors\)](#), show that the  $i$ th residual is  $N(0, \sigma^2(1 - m_{ii}))$ , where  $m_{ii}$  is the  $i$ th diagonal entry of  $\mathbf{M}$ , and that it is zero with probability one iff  $m_{ii} = 1$ .

**3.3.5.2 B. Practice****Exercise B1.**

Let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$  and let  $\mathbf{Q}$  be an orthogonal matrix whose first row is  $n^{-1/2}\mathbf{1}^\top$ . Show that the entries of  $\mathbf{V} = \mathbf{Q}\mathbf{Y}$  are independent, with  $V_1 \sim N(\sqrt{n}\mu, \sigma^2)$  and  $V_i \sim N(0, \sigma^2)$  for  $i \geq 2$ . Show that  $\sum_i (Y_i - \bar{Y})^2 = \sum_{i \geq 2} V_i^2$ , and conclude again that  $\bar{Y}$  and  $S^2$  are independent.

**Exercise B2.**

Let  $Z \sim N(0, 1)$  and, for  $c > 0$ , let  $W = Z$  if  $|Z| \leq c$  and  $W = -Z$  otherwise. Show that  $W \sim N(0, 1)$ , that there is a  $c$  with  $\text{Cov}(Z, W) = 0$ , and that  $(Z, W)$  is not jointly normal for any  $c$ .

*Solution.* Because  $Z$  and  $-Z$  have the same distribution and the event  $\{|Z| > c\}$  is unchanged by the sign change,  $\Pr(W \leq w) = \Pr(Z \leq w, |Z| \leq c) + \Pr(-Z \leq w, |Z| > c) = \Pr(Z \leq w)$ . Next,  $\text{Cov}(Z, W) = E(Z^2; |Z| \leq c) - E(Z^2; |Z| > c) = 2E(Z^2; |Z| \leq c) - 1$ , which is continuous and increasing in  $c$ , tends to  $-1$  as  $c \rightarrow 0$  and to  $1$  as  $c \rightarrow \infty$ , and so vanishes at some  $c$ . Finally  $Z + W = 2Z\mathbf{1}_{\{|Z| \leq c\}}$  equals 0 with probability  $\Pr(|Z| > c) \in (0, 1)$  but is not constant. A normal variable is either constant or has a density, so  $Z + W$  is not normal.

**Exercise B3.**

For the pair  $(Z, W)$  of [Example \(A random sign\)](#), find every  $\mathbf{a} \neq \mathbf{0}$  for which  $a_1Z + a_2W$  is normal. *Hint:* an equal mixture of  $N(0, v_1)$  and  $N(0, v_2)$  has fourth moment  $\frac{3}{2}(v_1^2 + v_2^2)$ . Conclude that  $(Z, W)$  is normal along exactly two lines through the origin, the coordinate axes.